

User Manual

for S32 GPT Driver

Document Number: UM11GPTASR4.4 Rev0000R4.0.2 Rev. 1.0

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Chapter 1

Revision History

Revision	Date	Author	Description
1.0	30.06.2023	NXP RTD Team	S32 Real-Time Drivers AUTOSAR 4.4 Version 4.0.2 Release

Chapter 2

Introduction

- [Supported Derivatives](#)
- [Overview](#)
- [About This Manual](#)
- [Acronyms and Definitions](#)
- [Reference List](#)

This User Manual describes NXP Semiconductor AUTOSAR GPT for S32. AUTOSAR GPT driver configuration parameters and deviations from the specification are described in GPT Driver chapter of this document. AUTOSAR GPT driver requirements and APIs are described in the AUTOSAR GPT driver software specification document.

2.1 Supported Derivatives

The software described in this document is intended to be used with the following microcontroller devices of NXP Semiconductors:

- s32g274a_bga525
- s32g254a_bga525
- s32g233a_bga525
- s32g234m_bga525
- s32g378a_bga525
- s32g379a_bga525
- s32g398a_bga525
- s32g399a_bga525
- s32g338m_bga525
- s32g339m_bga525
- s32g358a_bga525
- s32g359a_bga525
- s32r45_bga780

All of the above microcontroller devices are collectively named as S32.

2.2 Overview

AUTOSAR (AUTomotive Open System ARchitecture) is an industry partnership working to establish standards for software interfaces and software modules for automobile electronic control systems.

AUTOSAR:

- paves the way for innovative electronic systems that further improve performance, safety and environmental friendliness.
- is a strong global partnership that creates one common standard: "Cooperate on standards, compete on implementation".
- is a key enabling technology to manage the growing electrics/electronics complexity. It aims to be prepared for the upcoming technologies and to improve cost-efficiency without making any compromise with respect to quality.
- facilitates the exchange and update of software and hardware over the service life of the vehicle.

2.3 About This Manual

This Technical Reference employs the following typographical conventions:

- **Boldface** style: Used for important terms, notes and warnings.
- *Italic* style: Used for code snippets in the text. Note that C language modifiers such "const" or "volatile" are sometimes omitted to improve readability of the presented code.

Notes and warnings are shown as below:

Note

This is a note.

Warning

This is a warning

2.4 Acronyms and Definitions

Term	Definition
API	Application Programming Interface
ASM	Assembler
BSMI	Basic Software Make file Interface
GPT	General Purpose Timer
C/CPP	C and C++ Source Code
CS	Chip Select
CTU	Cross Trigger Unit
DEM	Diagnostic Event Manager
DET	Development Error Tracer
DMA	Direct Memory Access
ECU	Electronic Control Unit
FIFO	First In First Out
LSB	Least Significant Bit
MCU	Micro Controller Unit
MIDE	Multi Integrated Development Environment
MSB	Most Significant Bit
N/A	Not Applicable
RAM	Random Access Memory
SIU	Systems Integration Unit
SWS	Software Specification
VLE	Variable Length Encoding
XML	Extensible Markup Language

2.5 Reference List

#	Title	Version
1	Specification of GPT Driver	AUTOSAR Release 4.4.0
2	S32G2 Reference Manual	Rev. 7, February 2023
3	S32G3 Reference Manual	Rev. 3, Feb 2023
4	S32R45 Reference Manual	Rev. 3, 12/2021
5	S32G2 Errata Document	Mask Set Errata for Mask 0P77B Rev. 3.0
6	S32G2 Errata Document	Mask Set Errata for Mask 1P77B Rev. 1.0
7	S32G3 Errata Document	Mask Set Errata for Mask 1P72B Rev. 2.1
8	S32R45 Errata Document	Mask Set Errata for Mask P57D, Rev. 2.0
9	S32G2 Data Sheet	Rev. 6, Dec 2022
10	S32G3 Data Sheet	Rev. 2, RC, Feb 2023
11	VR5510 Data Sheet	Rev. 5, April 2022
12	S32R45 Data Sheet	Rev. 3, RC — 11/2022

Chapter 3

Driver

- [Requirements](#)
- [Driver Design Summary](#)
- [Hardware Resources](#)
- [Deviations from Requirements](#)
- [Driver Limitations](#)
- [Driver usage and configuration tips](#)
- [Runtime errors](#)
- [Symbolic Names Disclaimer](#)

3.1 Requirements

- Requirements for this driver are detailed in the RTD GPT AUTOSAR Release 4.4.0.
- Driver Software Specification document (See S32G3 Reference Manual, Rev. 2, June 2022, S32G2 Reference Manual, Rev. 5, May 2022 and S32R45 Reference Manual, Rev. 3, 12/2021).

3.2 Driver Design Summary

The RTD driver assures reentrancy (single core execution) for the APIs based on the following assumptions:

- The "called-again" API is for a different resource (hardware/logic channel);
- Common variables/registers accessed with "rmw" are guarded by Exclusive Areas which need to be correctly implemented in RTE on user side;

The GPT Driver implements the following channels on S32 peripherals.

The table provides information regarding the Timer channels available for the various derivatives across different packages in S32G274 family. This table lists only the supported packages by GPT driver.

RTC module features :

- 32-bit counter
- RTC interrupt with interrupt enable.
- Selectable counter clock sources
- Counter runs in all modes of operation.

PIT timer module features :

- Four 32-bit counters per module
- Independent timeout periods for each timer
- Independent interrupt source.

STM timer module features :

- One 32-bit up counter with 8-bit prescaler (1 to 256) per module.
- Four 32-bit compare channels
- Independent interrupt source for each channel.

FTM timer module features :

- One 16-bit up counter with 16-bit
- Prescaler divide-by 1, 2, 4, 8, 16, 32, 64, or 128
- Independent interrupt source for each channel.

3.3 Hardware Resources

#	Hardware IP	Description
1	STM	System Timer Module
2	PIT	Periodic Interrupt Timer
3	RTC	Real Time Clock
4	FTM	FlexTimer

3.4 Deviations from Requirements

The driver deviates from the AUTOSAR GPT Driver software specification in some places.

There are also some additional requirements (on top of requirements detailed in AUTOSAR GPT Driver software specification) which need to be satisfied for correct operation.

Deviations Status Column Description

Term	Definition
N/S	Out of scope
N/I	Not implemented
N/F	Not fully implemented

Below table identifies the AUTOSAR requirements that are not fully implemented, implemented differently, or out of scope for the driver.

Requirement	Status	Description	Notes
SWS_Gpt_00261	N/S	Gpt_Irq.c shall include Gpt.h for the prototype declaration of the notification functions.	Rejection reason: Gpt_Irq.c is not needed. Autosar specific interrupt behaviour is implemented using a normal function placed in the Gpt.c file.
SWS_Gpt_00278	N/S	Module - Header File - Imported Type - EcuM_flex - EcuM.h - EcuM_WakeupSourceType - Std_Types - StandardTypes.h - Std_ReturnType - StandardTypes.h - Std_VersionInfoType -	Rejection reason: No production errors needed for current development.
SWS_Gpt_00381	N/S	These requirements are not applicable to this specification.	Not a requirement
ECUC_Gpt_00235	N/S	Container Name - GptWakeupConfiguration - Description - Function pointer to callback function (for wakeup notification). - Configuration Parameters -	Rejection reason: Wrong Description: Function pointer to callback function (for non-wakeup notification). It shall relate to wakeup configuration.
SWS_Gpt_CONSTR_00001	N/S	DRAFT: The ECUC partitions referenced by GptKernelEcucPartitionRef shall be a subset of the ECUC partitions referenced by GptEcucPartitionRef.()	Type IV Autosar multicore not implemented for current module. AAI-445; Agree that each module can reject the Autosar Standard requirement

Requirement	Status	Description	Notes
ECUC_Gpt_00338	N/S	Name - GptKernelEcuc↔ PartitionRef - Parent Container - GptDriverConfiguration - Description - Maps the GPT kernel to zero or one ECUC partitions to assign the driver kernel to a certain core. The ECUC partition referenced is a subset of the ECUC partitions where the GPT driver is mapped to. Note: The kernel reference shall not be set in case the GPT driver is implemented without a kernel (refer to definition of GptEcucPartitionRef).Tags: atp.Status=draft - Multiplicity - 0..1 - Type - Reference to [EcucPartition] - Post-Build Variant Multiplicity - true - Post-Build Variant Value - true - Multiplicity Configuration Class - Pre-compile time - X - All Variants - Link time - - - - Post-build time - - - - Value Configuration Class - Pre-compile time - X - All Variants - Link time - - - - Post-build time - - - - Scope / Dependency - scope: ECU -	Type IV Autosar multicore not implemented for current module. AAI-445; Agree that each module can reject the Autosar Standard requirement.

3.5 Driver Limitations

The GPT driver software have some following limitations for RTD S32 RTM :

- Multicore aware support not full validated.
- S32DS not support multiple configurations.

3.6 Driver usage and configuration tips

In this chapter, the extra features from our drivers that are not described in the AutoSAR standard are detailed.

3.7 Runtime errors

The driver generates the following DEM errors at runtime.

Function	Error Code	Condition triggering the error
Gpt_ValidateChannelStatus()	GPT_E_BUSY	API service called when timer channel is still busy (running)
Gpt_ValidateMode()	GPT_E_MODE	API service called when driver is in wrong mode

3.8 Symbolic Names Disclaimer

All containers having symbolicNameValue set to TRUE in the AUTOSAR schema will generate defines like:

```
#define <Mip>Conf_<Container_ShortName>_<Container_ID>
```

For this reason it is forbidden to duplicate the names of such containers across the RTD configurations or to use names that may trigger other compile issues (e.g. match existing `#ifdefs` arguments).

Chapter 4

Tresos Configuration Plug-in

This chapter describes the Tresos configuration plug-in for the driver. All the parameters are described below.

- Module [Gpt](#)
 - Container [GptChannelConfigSet](#)
 - * Container [GptChannelConfiguration](#)
 - Parameter [GptChannelId](#)
 - Parameter [GptHwIp](#)
 - Parameter [GptChannelMode](#)
 - Parameter [GptChannelTickFrequency](#)
 - Parameter [GptChannelTickValueMax](#)
 - Parameter [GptEnableWakeup](#)
 - Parameter [GptNotification](#)
 - Reference [GptChannelEcucPartitionRef](#)
 - Reference [GptModuleRef](#)
 - Reference [GptChannelClkSrcRef](#)
 - Container [GptWakeupConfiguration](#)
 - Reference [GptWakeupSourceRef](#)
 - * Container [GptPit](#)
 - Parameter [GptPitModule](#)
 - Parameter [PitFreezeEnable](#)
 - Container [GptPitChannels](#)
 - Parameter [GptPitChannel](#)
 - Parameter [ChainMode](#)
 - * Container [GptStm](#)
 - Parameter [GptStmModule](#)
 - Parameter [GptStmPrescaler](#)
 - Parameter [GptStmAlternatePrescaler](#)
 - Parameter [StmFreezeEnable](#)
 - Container [GptStmChannels](#)
 - Parameter [GptStmChannel](#)
 - Parameter [StmAbsoluteCounting](#)

- * Container [GptRtc](#)
 - Parameter [GptRtcModule](#)
 - Parameter [RtcFreezeEnable](#)
 - Parameter [DivBy512](#)
 - Parameter [DivBy32](#)
 - Parameter [GptRtcChannelClkSrc](#)
- * Container [GptFtm](#)
 - Parameter [GptFtmModule](#)
 - Parameter [GptFtmPrescaler](#)
 - Parameter [GptFtmAlternatePrescaler](#)
 - Parameter [GptFtmChannelClkSrc](#)
 - Parameter [FtmFreezeEnable](#)
 - Parameter [GptFtmCountingMode](#)
 - Container [GptFtmChannels](#)
 - Parameter [GptFtmChannel](#)
 - Parameter [FtmAbsoluteCounting](#)
- Container [GptHwConfiguration](#)
 - * Parameter [GptIsrHwId](#)
 - * Parameter [GptIsrEnable](#)
 - * Parameter [GptChannelsUsed](#)
- Container [GptConfigurationOfOptApiServices](#)
 - * Parameter [GptDeinitApi](#)
 - * Parameter [GptEnableDisableNotificationApi](#)
 - * Parameter [GptTimeElapsedApi](#)
 - * Parameter [GptTimeRemainingApi](#)
 - * Parameter [GptVersionInfoApi](#)
 - * Parameter [GptWakeupFunctionalityApi](#)
 - * Parameter [GptPredefTimerFunctionalityApi](#)
- Container [GptAutosarExt](#)
 - * Parameter [GptEnableDualClockMode](#)
 - * Parameter [GptChangeNextTimeoutValueApi](#)
 - * Parameter [GptEnableUserModeSupport](#)
 - * Parameter [ChainModeApi](#)
 - * Parameter [GptStandbyWakeupSupport](#)
- Container [GptDriverConfiguration](#)
 - * Parameter [GptDevErrorDetect](#)

- * Parameter [GptPredefTimer100us32bitEnable](#)
- * Parameter [GptMulticoreSupport](#)
- * Parameter [GptPredefTimer1usEnablingGrade](#)
- * Parameter [GptTimeoutMethod](#)
- * Parameter [GptTimeoutDuration](#)
- * Parameter [GptReportWakeupSource](#)
- * Reference [GptEcucPartitionRef](#)
- * Reference [GptKernelEcucPartitionRef](#)
- * Container [GptClockReferencePoint](#)
 - Reference [GptClockReference](#)
- Container [GptPredefTimerConfiguration](#)
 - * Container [GptPredefTimer_1us_16Bit](#)
 - Parameter [GptHwChannel](#)
 - Parameter [GptFtmChannelClkSrc](#)
 - Parameter [GptChannelPrescaler](#)
 - Parameter [GptFreezeEnable](#)
 - Reference [GptChannelClkSrcRef](#)
 - * Container [GptPredefTimer_1us_24Bit](#)
 - Parameter [GptHwChannel](#)
 - Parameter [GptChannelPrescaler](#)
 - Parameter [GptFreezeEnable](#)
 - Reference [GptChannelClkSrcRef](#)
 - * Container [GptPredefTimer_1us_32Bit](#)
 - Parameter [GptHwChannel](#)
 - Parameter [GptChannelPrescaler](#)
 - Parameter [GptFreezeEnable](#)
 - Reference [GptChannelClkSrcRef](#)
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 - Parameter [GptChannelPrescaler](#)
 - Parameter [GptFreezeEnable](#)
 - Reference [GptChannelClkSrcRef](#)
- Container [CommonPublishedInformation](#)
 - * Parameter [ArReleaseMajorVersion](#)
 - * Parameter [ArReleaseMinorVersion](#)
 - * Parameter [ArReleaseRevisionVersion](#)
 - * Parameter [ModuleId](#)
 - * Parameter [SwMajorVersion](#)
 - * Parameter [SwMinorVersion](#)
 - * Parameter [SwPatchVersion](#)
 - * Parameter [VendorApiInfix](#)
 - * Parameter [VendorId](#)

4.1 Module Gpt

Configuration of the Gpt (General Purpose Timer) module.

Included containers:

- [GptChannelConfigSet](#)
- [GptHwConfiguration](#)
- [GptConfigurationOfOptApiServices](#)
- [GptAutosarExt](#)
- [GptDriverConfiguration](#)
- [GptPredefTimerConfiguration](#)
- [CommonPublishedInformation](#)

Property	Value
type	ECUC-MODULE-DEF
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantSupport	true
supportedConfigVariants	VARIANT-POST-BUILD, VARIANT-PRE-COMPILE

4.2 Container GptChannelConfigSet

This container is the base of an Configuration Set which contains the configured GPT channels.

This way, different configuration sets can be defined for post-build process.

Included subcontainers:

- [GptChannelConfiguration](#)
- [GptPit](#)
- [GptStm](#)
- [GptRtc](#)
- [GptFtm](#)

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.3 Container GptChannelConfiguration

This container contains the channel-wide configuration (parameters) of the GPT Driver

Included subcontainers:

- [GptWakeupConfiguration](#)

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	Infinite
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE

4.4 Parameter GptChannelId

Channel Id of the GPT channel. This value will be assigned to the symbolic name derived of the GptChannelConfiguration container short name.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	true
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	0
max	4294967295
min	0

4.5 Parameter GptHwIp

Vendor specific: Selects the physical GPT Channel.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
default Value	STM
literals	['STM', 'PIT', 'FTM', 'RTC']

4.6 Parameter GptChannelMode

Specifies the behaviour of the timerchannel after the timeout has expired

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
default Value	GPT_CH_MODE_ONESHOT
literals	['GPT_CH_MODE_CONTINUOUS', 'GPT_CH_MODE_ONESHOT']

4.7 Parameter GptChannelTickFrequency

EN: Specifies the tick frequency of the timer channel in Hz.

Property	Value
type	ECUC-FLOAT-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false

Property	Value
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	0.0
max	1.6E8
min	0.0

4.8 Parameter GptChannelTickValueMax

Maximum value in ticks, the timer channel is able to count. With the next tick, the timer rolls over to zero.

It is mandatory to set 4294967295 for the PIT, RTC and STM channels (coresponding to the 32 bits counter resolution)

and 65535 for the FTM channels (corresponding to the 16 bits counter resolution)!

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	4294967295
max	4294967295
min	65535

4.9 Parameter GptEnableWakeup

Enables wakeup capability of CPU for a channel.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF

Property	Value
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.10 Parameter GptNotification

Function pointer to callback function(for non-wakeup notification).

The field is editable only if the switch GptEnableDisableNotificationApi is true.

Property	Value
type	ECUC-FUNCTION-NAME-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: POST-BUILD
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: POST-BUILD
defaultValue	NULL_PTR

4.11 Reference GptChannelEcucPartitionRef

Maps a GPT channel to zero or one ECUC partition to limit the access to this channel group. The ECUC partitions referenced are a subset of the ECUC partitions where the GPT driver is mapped to.

Property	Value
type	ECUC-REFERENCE-DEF
origin	AUTOSAR_ECUC
lowerMultiplicity	0

Property	Value
upperMultiplicity	Infinite
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	true
valueConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	False
destination	/AUTOSAR/EcucDefs/EcuC/EcucPartitionCollection/EcucPartition

4.12 Reference GptModuleRef

Maps a GPT channel to zero or one ECUC partition to limit the access to this channel group. The ECUC partitions referenced are a subset of the ECUC partitions where the GPT driver is mapped to.

Property	Value
type	ECUC-CHOICE-REFERENCE-DEF
origin	NXP
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
requiresSymbolicNameValue	False
destinations	['/TS_T40D11M40I2R0/Gpt/GptChannelConfigSet/GptPit/GptPitChannels', '/TS_T40D11M40I2R0/Gpt/GptChannelConfigSet/GptStm/GptStmChannels', '/TS_T40D11M40I2R0/Gpt/GptChannelConfigSet/GptRtc', '/TS_T40D11M40I2R0/Gpt/GptChannelConfigSet/GptFtm/GptFtmChannels']

4.13 Reference GptChannelClkSrcRef

Reference to the GptClockReferencePoint from which the channel clock is derived.

Property	Value
type	ECUC-REFERENCE-DEF
origin	AUTOSAR_ECUC
lowerMultiplicity	1
upperMultiplicity	1

Property	Value
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
requiresSymbolicNameValue	False
destination	/AUTOSAR/EcuDefs/Gpt/GptDriverConfiguration/GptClockReferencePoint

4.14 Container GptWakeupConfiguration

This container defines the wakeup source codes reported to Ecu State Manager.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE

4.15 Reference GptWakeupSourceRef

In case the wakeup-capability is true this value is transmitted to the Ecu State Manager.

Implementation Type: reference to EcuM_WakeupSourceType

Property	Value
type	ECUC-REFERENCE-DEF
origin	AUTOSAR_ECUC
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE

Property	Value
	VARIANT-POST-BUILD: POST-BUILD
requiresSymbolicNameValue	true
destination	/AUTOSAR/EcuDefs/EcuM/EcuMConfiguration/EcuMCommon↔ Configuration/EcuMWakeupSource

4.16 Container GptPit

Configuration of a Pit module available on the platform.

Included subcontainers:

- [GptPitChannels](#)

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	0
upperMultiplicity	2
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE

4.17 Parameter GptPitModule

Select the physical Pit Module.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	PIT_0
literals	['PIT_0', 'PIT_1']

4.18 Parameter PitFreezeEnable

Enables/Disables freeze bit.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
default Value	false

4.19 Container GptPitChannels

Pit hw channels.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	7
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE

4.20 Parameter GptPitChannel

Selects one of the Pit channels available on the platform.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP

Property	Value
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	CH_0
literals	['CH_0', 'CH_1', 'CH_2', 'CH_3', 'CH_4', 'CH_5', 'CH_6']

4.21 Parameter ChainMode

Enables/Disables chain mode

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.22 Container GptStm

Configuration of a Stm module available on the platform.

Included subcontainers:

- [GptStmChannels](#)

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	0

Property	Value
upperMultiplicity	13
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE

4.23 Parameter GptStmModule

Select the physical Stm Module.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	STM_0
literals	['STM_0', 'STM_1', 'STM_2', 'STM_3', 'STM_4', 'STM_5', 'STM_6', 'STM_7', 'STM_8', 'STM_9', 'STM_10', 'STM_11', 'STM_12']

4.24 Parameter GptStmPrescaler

Vendor specific: The GPT module specific clock prescaler value.

Note with STM:

- STM prescaler should be between 1-256.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

Property	Value
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	1
max	256
min	1

4.25 Parameter GptStmAlternatePrescaler

Vendor specific: The GPT module specific clock prescaler value.

Note with STM:

- STM prescaler should be between 1-256.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	1
max	256
min	1

4.26 Parameter StmFreezeEnable

Enables/Disables freeze bit.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1

Property	Value
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.27 Container GptStmChannels

STM hw channels

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	4
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE

4.28 Parameter GptStmChannel

Selects one of the Stm channels available on the platform.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	CH_0
literals	['CH_0', 'CH_1', 'CH_2', 'CH_3']

4.29 Parameter StmAbsoluteCounting

Enables/Disables absolute compare value.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
default Value	false

4.30 Container GptRtc

Configuration of a Pit module available on the platform.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	0
upperMultiplicity	2
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE

4.31 Parameter GptRtcModule

Select the physical Pit Module.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP

Property	Value
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: POST-BUILD
defaultValue	RTC_0_CH_0
literals	['RTC_0_CH_0']

4.32 Parameter RtcFreezeEnable

Enables/Disables freeze bit

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.33 Parameter DivBy512

Optional 512 prescaler

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A

Property	Value
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.34 Parameter DivBy32

Optional 32 prescaler

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.35 Parameter GptRtcChannelClkSrc

Selectable counter clock sources (IRCs and OSCs)

? Clock source 0

? Clock source 1 - reserved

? Clock source 2

? Clock source 3

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1

Property	Value
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	RTC_IP_CLOCK_SOURCE_0
literals	['RTC_IP_CLOCK_SOURCE_0', 'RTC_IP_CLOCK_SOURCE_2', 'RTC_IP_CLOCK_SOURCE_3']

4.36 Container GptFtm

Configuration of a Ftm module available on the platform.

Included subcontainers:

- [GptFtmChannels](#)

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	0
upperMultiplicity	2
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE

4.37 Parameter GptFtmModule

Select the physical Ftm Module.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE

Property	Value
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	FTM_0
literals	['FTM_0', 'FTM_1']

4.38 Parameter GptFtmPrescaler

Vendor specific: The GPT module specific clock prescaler value.

Note with FTM:

- FTM prescaler should be 1, 2, 4, 8, 16, 32, 64 or 128.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	1
max	128
min	1

4.39 Parameter GptFtmAlternatePrescaler

Vendor specific: The GPT module specific clock prescaler value.

Selects one of 8 division factors for the clock source selected by GptFtmChannelClkSrcRef. The new prescaler factor affects

the clock source on the next system clock cycle after the new value is updated into the register bits.

- 1 - Value written in register: 0
- 2 - Value written in register: 1
- 4 - Value written in register: 2

8 - Value written in register: 3

16 - Value written in register: 4

32 - Value written in register: 5

64 - Value written in register: 6

128 - Value written in register: 7

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	1
max	128
min	1

4.40 Parameter GptFtmChannelClkSrc

Vendor specific: The GPT module specific clock input for the timer unit can statically be configured and allows to select different clock sources per module.

Select the clock source for the FlexTimer module for this platform.

FTM_GPT_IP_CLOCK_SOURCE_NONENone use clock for FTM?

FTM_GPT_IP_CLOCK_SOURCE_SYSTEMCLKSystem clock

FTM_GPT_IP_CLOCK_SOURCE_FIXED_FREQUENCYFixed frequency clock

FTM_GPT_IP_CLOCK_SOURCE_EXTERNALCLKExternal clock

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1

Property	Value
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	FTM_GPT_IP_CLOCK_SOURCE_NONE
literals	['FTM_GPT_IP_CLOCK_SOURCE_NONE', 'FTM_GPT_IP_CLOCK_SOURCE_SYSTEMCLK', 'FTM_GPT_IP_CLOCK_SOURCE_FIXED_FREQUENCY', 'FTM_GPT_IP_CLOCK_SOURCE_EXTERNALCLK']

4.41 Parameter FtmFreezeEnable

Enables/Disables freeze bit.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.42 Parameter GptFtmCountingMode

This parameter indicates the count direction for the whole GPT driver.

Note This feature is not supported in current implementation. Up-counting is always used as default

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A

Property	Value
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	FTM_GPT_IP_MODE_UP_TIMER
literals	['FTM_GPT_IP_MODE_DOWN_TIMER', 'FTM_GPT_IP_MODE_UP_TIMER']

4.43 Container GptFtmChannels

Ftm hw channels

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	6
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE

4.44 Parameter GptFtmChannel

Selects one of the Ftm channels available on the platform.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD

Property	Value
defaultValue	CH_0
literals	['CH_0', 'CH_1', 'CH_2', 'CH_3', 'CH_4', 'CH_5']

4.45 Parameter FtmAbsoluteCounting

Enables/Disables absolute compare value.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	false

4.46 Container GptHwConfiguration

List of all HW channel resources for GPT module.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	88
upperMultiplicity	88
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.47 Parameter GptIsrHwId

ID of HW interrupt resources.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	true
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	STM_0_CH_1
literals	['STM_0_CH_0', 'STM_0_CH_1', 'STM_0_CH_2', 'STM_0_CH_3', 'STM_1_CH_0', 'STM_1_CH_1', 'STM_1_CH_2', 'STM_1_CH_3', 'STM_2_CH_0', 'STM_2_CH_1', 'STM_2_CH_2', 'STM_2_CH_3', 'STM_3_CH_0', 'STM_3_CH_1', 'STM_3_CH_2', 'STM_3_CH_3', 'STM_4_CH_0', 'STM_4_CH_1', 'STM_4_CH_2', 'STM_4_CH_3', 'STM_5_CH_0', 'STM_5_CH_1', 'STM_5_CH_2', 'STM_5_CH_3', 'STM_6_CH_0', 'STM_6_CH_1', 'STM_6_CH_2', 'STM_6_CH_3', 'STM_7_CH_0', 'STM_7_CH_1', 'STM_7_CH_2', 'STM_7_CH_3', 'STM_8_CH_0', 'STM_8_CH_1', 'STM_8_CH_2', 'STM_8_CH_3', 'STM_9_CH_0', 'STM_9_CH_1', 'STM_9_CH_2', 'STM_9_CH_3', 'STM_10_CH_0', 'STM_10_CH_1', 'STM_10_CH_2', 'STM_10_CH_3', 'STM_11_CH_0', 'STM_11_CH_1', 'STM_11_CH_2', 'STM_11_CH_3', 'STM_12_CH_0', 'STM_12_CH_1', 'STM_12_CH_2', 'STM_12_CH_3', 'PIT_0_CH_0', 'PIT_0_CH_1', 'PIT_0_CH_2', 'PIT_0_CH_3', 'PIT_0_CH_4', 'PIT_0_CH_5', 'PIT_0_CH_6', 'PIT_1_CH_0', 'PIT_1_CH_1', 'PIT_1_CH_2', 'PIT_1_CH_3', 'PIT_1_CH_4', 'PIT_1_CH_5', 'FTM_0_CH_0', 'FTM_0_CH_1', 'FTM_0_CH_2', 'FTM_0_CH_3', 'FTM_0_CH_4', 'FTM_0_CH_5', 'FTM_1_CH_0', 'FTM_1_CH_1', 'FTM_1_CH_2', 'FTM_1_CH_3', 'FTM_1_CH_4', 'FTM_1_CH_5', 'RTC_0_CH_0', 'STM_0_PREDEF', 'STM_1_PREDEF', 'STM_2_PREDEF', 'STM_3_PREDEF', 'STM_4_PREDEF', 'STM_5_PREDEF', 'STM_6_PREDEF', 'STM_7_PREDEF', 'FTM_0_PREDEF', 'FTM_1_PREDEF']

4.48 Parameter GptIsrEnable

Enable/Disable HW channels' Interrupt Sources.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP

Property	Value
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.49 Parameter GptChannelsUsed

This column configures HW channels which are going to be used.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.50 Container GptConfigurationOfOptApiServices

This container contains all configuration switches for configuring optional API services of the GPT driver.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	S32GPT Driver

4.51 Parameter GptDeinitApi

Adds / removes the service Gpt_DeInit() from the code.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	true

4.52 Parameter GptEnableDisableNotificationApi

Adds / removes the services Gpt_EnableNotification() and Gpt_DisableNotification from the code.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	true

4.53 Parameter GptTimeElapsedApi

Adds / removes the service Gpt_GetTimeElapsed() from the code.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF

Property	Value
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	true

4.54 Parameter GptTimeRemainingApi

Adds / removes the service Gpt_GetTimeRemaining() from the code.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	true

4.55 Parameter GptVersionInfoApi

Adds / removes the service Gpt_GetVersionInfo() from the code.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A

Property	Value
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
default Value	false

4.56 Parameter GptWakeupFunctionalityApi

Adds / removes the services Gpt_SetMode(), Gpt_EnableWakeup() Gpt_DisableWakeup() and Gpt_Cbk_CheckWakeup() from the code.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
default Value	true

4.57 Parameter GptPredefTimerFunctionalityApi

Adds / removes the services Gpt_SetMode(), Gpt_EnableWakeup() Gpt_DisableWakeup() and Gpt_Cbk_CheckWakeup() from the code.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
default Value	true

4.58 Container GptAutosarExt

Enabling the settings of this section will configure the driver in a mode not compliant with AUTOSAR requirements.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.59 Parameter GptEnableDualClockMode

Enables prescaler settings at mode transition.true: Enabled.false: Disabled.

Note This feature is not required by Autosar.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	false

4.60 Parameter GptChangeNextTimeoutValueApi

Vendor specific: Enables settings for changing the channel counter compare value of a running counter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF

Property	Value
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	false

4.61 Parameter GptEnableUserModeSupport

When this parameter is enabled, the GPT module will adapt to run from User Mode. There is no difference between User mode and Privileged mode in GPT module.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	false

4.62 Parameter ChainModeApi

Vendor specific: Enable/disable API for Chain Mode support.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false

Property	Value
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	false

4.63 Parameter GptStandbyWakeupSupport

The driver shall NOT CLEAR the interrupt flag, the interrupt enable bit and also should not disable the counter, during init (Gpt_SRtc_Init()) the flag is already set.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	false

4.64 Container GptDriverConfiguration

This container contains the module-wide configuration (parameters) of the GPT Driver.

Included subcontainers:

- [GptClockReferencePoint](#)

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.65 Parameter GptDevErrorDetect

Enables/Disables development error detection.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	false

4.66 Parameter GptPredefTimer100us32bitEnable

Enables/Disables the feature 100us/ tick

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.67 Parameter GptMulticoreSupport

Enables/Disables Multicore Support.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF

Property	Value
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	false

4.68 Parameter GptPredefTimer1usEnablingGrade

Specifies the grade of enabling the GPT Predef Timers with 1us tick duration.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	GPT_PREDEF_TIMER_1US_16BIT_ENABLED
literals	['GPT_PREDEF_TIMER_1US_16BIT_ENABLED', 'GPT_PREDEF_TIMER_1US_16_24BIT_ENABLED', 'GPT_PREDEF_TIMER_1US_16_24_32BIT_ENABLED', 'GPT_PREDEF_TIMER_1US_DISABLED']

4.69 Parameter GptTimeoutMethod

GptTimeoutMethod: Configures the timeout method.

Based on this selection a certain timeout method from OsIf will be used in the driver.

Note: If SystemTimer or CustomTimer are selected make sure the corresponding timer is enabled in OsIf General configuration.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	OSIF_COUNTER_DUMMY
literals	['OSIF_COUNTER_SYSTEM', 'OSIF_COUNTER_CUSTOM', 'OSIF_COUNTER_DUMMY']

4.70 Parameter GptTimeoutDuration

The unit of measurement is given in number of microseconds. This is a timeout value which is used to wait till - PIT_RTI_LDVAL is synchronized into the RTI clock domain. If the Status is not updated then after this timeout a runtime error will be reported. This parameter is used for PitRti only.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	800
max	65535
min	1

4.71 Parameter GptReportWakeupSource

Enables/Disables wakeup source reporting.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	False
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
default Value	true

4.72 Reference GptEcucPartitionRef

Maps the GPT driver to zero or multiple ECUC partitions to make the driver API available in the according partition. Depending on the addressed timer resource the interfaces operate as follows:

In case of partition local timer resources (n:1 mapping) the API operates as an independent instance in the according ECUC partition.

In case of global timer resources (1:m mapping) the API operates on the global timer resource either by protected access to the resource or by implementing an according kernel.

Property	Value
type	ECUC-REFERENCE-DEF
origin	AUTOSAR_ECUC
lowerMultiplicity	0
upperMultiplicity	Infinite
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	true
valueConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	False
destination	/AUTOSAR/EcucDefs/EcuC/EcucPartitionCollection/EcucPartition

4.73 Reference GptKernelEcucPartitionRef

Maps the GPT kernel to zero or one ECUC partitions to assign the driver kernel to a certain core. The ECUC partition referenced is a subset of the ECUC partitions where the GPT driver is mapped to.

Property	Value
type	ECUC-REFERENCE-DEF
origin	AUTOSAR_ECUC
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	true
valueConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	False
destination	/AUTOSAR/EcuDefs/EcuC/EcuPartitionCollection/EcuPartition

4.74 Container GptClockReferencePoint

This container contains a parameter, which represents a reference to a container of the type McuClockReferencePoint (defined in module MCU).

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	Infinite
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE

4.75 Reference GptClockReference

Reference to a container of the type McuClockReferencePoint, to select an input clock.

Property	Value
type	ECUC-REFERENCE-DEF
origin	AUTOSAR_ECUC
lowerMultiplicity	1
upperMultiplicity	1

Property	Value
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	False
destination	/AUTOSAR/EcuDefs/Mcu/McuModuleConfiguration/McuClockSetting↔ Config/McuClockReferencePoint

4.76 Container GptPredefTimerConfiguration

Container for configuring the Predefined Timer functionality.

Included subcontainers:

- [GptPredefTimer_1us_16Bit](#)
- [GptPredefTimer_1us_24Bit](#)
- [GptPredefTimer_1us_32Bit](#)
- [GptPredefTimer_100us_32Bit](#)

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.77 Container GptPredefTimer__1us__16Bit

This container contains the 1U_16BIT predef timer configuration (parameters) of the GPT Driver

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF

Property	Value
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE

4.78 Parameter GptHwChannel

Vendor specific: Selects the physical GPT Channel. PIT not use for this feature

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	STM_0_PREDEF
literals	['STM_0_PREDEF', 'STM_1_PREDEF', 'STM_2_PREDEF', 'STM_3_P← REDEF', 'STM_4_PREDEF', 'STM_5_PREDEF', 'STM_6_PREDEF', 'ST← M_7_PREDEF', 'FTM_0_PREDEF', 'FTM_1_PREDEF']

4.79 Parameter GptFtmChannelClkSrc

Vendor specific: The GPT module specific clock input for the timer unit can statically be configured and allows to select different clock sources per module.

Select the clock source for the FlexTimer module for this platform.

FTM_GPT_IP_CLOCK_SOURCE_NONENone use clock for FTM?

FTM_GPT_IP_CLOCK_SOURCE_SYSTEMCLKSystem clock

FTM_GPT_IP_CLOCK_SOURCE_FIXED_FREQUENCYFixed frequency clock

FTM_GPT_IP_CLOCK_SOURCE_EXTERNALCLKExternal clock

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	FTM_GPT_IP_CLOCK_SOURCE_NONE
literals	['FTM_GPT_IP_CLOCK_SOURCE_NONE', 'FTM_GPT_IP_CLOCK_SOURCE_SYSTEMCLK', 'FTM_GPT_IP_CLOCK_SOURCE_FIXED_FREQUENCY', 'FTM_GPT_IP_CLOCK_SOURCE_EXTERNALCLK']

4.80 Parameter GptChannelPrescaler

Vendor specific: The GPT module specific clock prescaler value.

Note with STM:

- STM prescaler should be between 1-256.

Note with FTM:

- FTM prescaler should be 1, 2, 4, 8, 16, 32, 64 or 128.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	1
max	256
min	1

4.81 Parameter GptFreezeEnable

Vendor specific: Select to set Freeze enable for the hw resources.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	true

4.82 Reference GptChannelClkSrcRef

Reference to the GptClockReferencePoint from which the channel clock is derived.

Property	Value
type	ECUC-REFERENCE-DEF
origin	NXP
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
requiresSymbolicNameValue	False
destination	/AUTOSAR/EcucDefs/Gpt/GptDriverConfiguration/GptClockReferencePoint

4.83 Container GptPredefTimer__1us__24Bit

This container contains the 1U_24BIT predef timer configuration (parameters) of the GPT Driver

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE

4.84 Parameter GptHwChannel

Vendor specific: Selects the physical GPT Channel.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	STM_0_PREDEF
literals	['STM_0_PREDEF', 'STM_1_PREDEF', 'STM_2_PREDEF', 'STM_3_P← REDEF', 'STM_4_PREDEF', 'STM_5_PREDEF', 'STM_6_PREDEF', 'ST← M_7_PREDEF']

4.85 Parameter GptChannelPrescaler

Vendor specific: The GPT module specific clock prescaler value.

Note with STM:

- STM prescaler should be between 1-256.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1

Property	Value
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	1
max	256
min	1

4.86 Parameter GptFreezeEnable

Vendor specific: Select to set Freeze enable for the hw resources.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	true

4.87 Reference GptChannelClkSrcRef

Reference to the GptClockReferencePoint from which the channel clock is derived.

Property	Value
type	ECUC-REFERENCE-DEF
origin	NXP
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true

Property	Value
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
requiresSymbolicNameValue	False
destination	/AUTOSAR/EcuDefs/Gpt/GptDriverConfiguration/GptClockReferencePoint

4.88 Container GptPredefTimer_1us_32Bit

This container contains the predef timer configuration (parameters) of the GPT Driver

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE

4.89 Parameter GptHwChannel

Vendor specific: Selects the physical GPT Channel.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	STM_0_PREDEF
literals	['STM_0_PREDEF', 'STM_1_PREDEF', 'STM_2_PREDEF', 'STM_3_P← REDEF', 'STM_4_PREDEF', 'STM_5_PREDEF', 'STM_6_PREDEF', 'ST← M_7_PREDEF']

4.90 Parameter GptChannelPrescaler

Vendor specific: The GPT module specific clock prescaler value.

Note with STM:

- STM prescaler should be between 1-256.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	1
max	256
min	1

4.91 Parameter GptFreezeEnable

Vendor specific: Select to set Freeze enable for the hw resources.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	true

4.92 Reference GptChannelClkSrcRef

Reference to the GptClockReferencePoint from which the channel clock is derived.

Property	Value
type	ECUC-REFERENCE-DEF
origin	NXP
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
requiresSymbolicNameValue	False
destination	/AUTOSAR/EcuDefs/Gpt/GptDriverConfiguration/GptClockReferencePoint

4.93 Container GptPredefTimer__100us__32Bit

This container contains the channel-wide configuration (parameters) of the GPT Driver

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE

4.94 Parameter GptHwChannel

Vendor specific: Selects the physical GPT Channel. PIT not use for this feature

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF

Property	Value
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	STM_0_PREDEF
literals	['STM_0_PREDEF', 'STM_1_PREDEF', 'STM_2_PREDEF', 'STM_3_P← REDEF', 'STM_4_PREDEF', 'STM_5_PREDEF', 'STM_6_PREDEF', 'ST← M_7_PREDEF']

4.95 Parameter GptChannelPrescaler

Vendor specific: The GPT module specific clock prescaler value.

Note with STM:

- STM prescaler should be between 1-256.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	1
max	256
min	1

4.96 Parameter GptFreezeEnable

Vendor specific: Select to set Freeze enable for the hw resources.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
default Value	true

4.97 Reference GptChannelClkSrcRef

Reference to the GptClockReferencePoint from which the channel clock is derived.

Property	Value
type	ECUC-REFERENCE-DEF
origin	NXP
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
requiresSymbolicNameValue	False
destination	/AUTOSAR/EcucDefs/Gpt/GptDriverConfiguration/GptClockReferencePoint

4.98 Container CommonPublishedInformation

Common container, aggregated by all modules. It contains published information about vendor and versions.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF

Property	Value
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.99 Parameter ArReleaseMajorVersion

Major version number of AUTOSAR specification on which the appropriate implementation is based on.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
defaultValue	4
max	4
min	4

4.100 Parameter ArReleaseMinorVersion

Minor version number of AUTOSAR specification on which the appropriate implementation is based on.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION

Property	Value
defaultValue	4
max	4
min	4

4.101 Parameter ArReleaseRevisionVersion

Revision version number of AUTOSAR specification on which the appropriate implementation is based on.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
defaultValue	0
max	0
min	0

4.102 Parameter ModuleId

Module ID of this module from Module List.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
defaultValue	100
max	100
min	100

4.103 Parameter SwMajorVersion

Major version number of the vendor specific implementation of the module. The numbering is vendor specific.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
defaultValue	4
max	4
min	4

4.104 Parameter SwMinorVersion

Minor version number of the vendor specific implementation of the module. The numbering is vendor specific.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
defaultValue	0
max	0
min	0

4.105 Parameter SwPatchVersion

Patch level version number of the vendor specific implementation of the module. The numbering is vendor specific.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
defaultValue	2
max	2
min	2

4.106 Parameter VendorApiInfix

In driver modules which can be instantiated several times on a single ECU, BSW00347 requires that the name of APIs is extended by the VendorId and a vendor specific name.

This parameter is used to specify the vendor specific name. In total, the implementation specific name is generated as follows:

<ModuleName>__>VendorId>__<VendorApiInfix>.

E.g. assuming that the VendorId of the implementor is 123 and the implementer chose a VendorApiInfix of "v11r456" a api name Can_Write defined in the SWS will translate to Can_123_v11r456Write.

This parameter is mandatory for all modules with upper multiplicity > 1. It shall not be used for modules with upper multiplicity =1.

Property	Value
type	ECUC-STRING-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
postBuildVariantValue	false
valueConfigClasses	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
defaultValue	

4.107 Parameter VendorId

Vendor ID of the dedicated implementation of this module according to the AUTOSAR vendor list.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
defaultValue	43
max	43
min	43



Chapter 5

Module Index

5.1 Software Specification

Here is a list of all modules:

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Chapter 6

Module Documentation

6.1 FTM IPL

6.1.1 Detailed Description

Data Structures

- struct [Ftm_Gpt_Ip_InstanceConfigType](#)
Structure to configure the FTM instance. [More...](#)
- struct [Ftm_Gpt_Ip_ChannelConfigType](#)
Structure to configure the FTM channels. [More...](#)
- struct [Ftm_Gpt_Ip_ChState](#)
internal context structure [More...](#)
- struct [Ftm_Gpt_Ip_InstancePrescalerType](#)
internal context structure [More...](#)

Macros

- `#define GPT\_STOP\_SEC\_CODE`
Ftm_Gpt_Ip_SetUserAccessAllowed.

Types Reference

- typedef void(* [Ftm_Gpt_Ip_CallbackType](#)) (uint8 callbackParam)
Callback type for each channel.

Enum Reference

- enum [Ftm_Gpt_Ip_CountingMode](#)
Unit options for counting mode.
- enum [Ftm_Gpt_Ip_ClockSource](#)
Enum containing the FTM module clock sources.
- enum [Ftm_Gpt_Ip_ChannelModeType](#)
Prescaler type. Indicates of whether the clock channel mode is "NORMAL" or "ALTERNATE".

6.1.2 Data Structure Documentation

6.1.2.1 struct Ftm_Gpt_Ip_InstanceConfigType

Structure to configure the FTM instance.

This structure holds the configuration settings for InstanceConfigType

Implements [Ftm_Gpt_Ip_InstanceConfigType](#)

Definition at line 166 of file Ftm_Gpt_Ip_Types.h.

Data Fields

Type	Name	Description
boolean	freezeBits	Enable/Disable freeze bits.
Ftm_Gpt_Ip_ClockSource	clocksource	Select FTM clocksource.
uint8	clockPrescaler	Select prescalerValue.
Ftm_Gpt_Ip_CountingMode	mode	Select mode.

6.1.2.2 struct Ftm_Gpt_Ip_ChannelConfigType

Structure to configure the FTM channels.

This structure holds the configuration settings for the ChannelConfigType

Implements [Ftm_Gpt_Ip_ChannelConfigType](#)

Definition at line 183 of file Ftm_Gpt_Ip_Types.h.

Data Fields

Type	Name	Description
uint8	hwChannel	hwChannel

Module Documentation

Data Fields

Type	Name	Description
Ftm_Gpt_Ip_CallbackType	callback	callback
uint8	callbackParam	callbackParam
Ftm_Gpt_Ip_ChannelModeType	channelMode	channelMode

6.1.2.3 struct Ftm_Gpt_Ip_ChState

internal context structure

This structure is used by the IPL driver for internal logic. The content is populated on InitChannel

Definition at line 197 of file Ftm_Gpt_Ip_Types.h.

Data Fields

Type	Name	Description
boolean	chInit	chInit
Ftm_Gpt_Ip_CallbackType	callback	callback
uint8	callbackParam	callbackParam
Ftm_Gpt_Ip_ChannelModeType	channelMode	channelMode

6.1.2.4 struct Ftm_Gpt_Ip_InstancePrescalerType

internal context structure

This structure is used by the IPL driver for internal logic. The content is populated on Init

Definition at line 211 of file Ftm_Gpt_Ip_Types.h.

Data Fields

Type	Name	Description
uint8	clockPrescaler	Clock divide value for the NormalPrescaler.
uint8	clockAlternatePrescaler	Clock divide value for the AlternatePrescaler.

6.1.3 Macro Definition Documentation

6.1.3.1 GPT_STOP_SEC_CODE

```
#define GPT_STOP_SEC_CODE
```

Ftm_Gpt_Ip_SetUserAccessAllowed.

This function is called externally by OS Application

Parameters

in	<i>FtmBaseAddr</i>	- The base address of PIT.
----	--------------------	----------------------------

Definition at line 118 of file Ftm_Gpt_Ip_TrustedFunctions.h.

6.1.4 Types Reference

6.1.4.1 Ftm_Gpt_Ip_CallbackType

```
typedef void(* Ftm_Gpt_Ip_CallbackType) (uint8 callbackParam)
```

Callback type for each channel.

Ftm_Gpt_Ip_CallbackType

Definition at line 158 of file Ftm_Gpt_Ip_Types.h.

6.1.5 Enum Reference

6.1.5.1 Ftm_Gpt_Ip_CountingMode

```
enum Ftm_Gpt_Ip_CountingMode
```

Unit options for counting mode.

This is used to choose how the timer is counting

Enumerator

FTM_GPT_IP_MODE_UP_TIMER	Timer with up counter.
FTM_GPT_IP_MODE_UP_DOWN_TIMER	timer with up-down counter

Definition at line 109 of file Ftm_Gpt_Ip_Types.h.

6.1.5.2 Ftm_Gpt_Ip_ClockSource

enum `Ftm_Gpt_Ip_ClockSource`

Enum containing the FTM module clock sources.

This is used to choose the FTM clock sources.

Enumerator

FTM_GPT_IP_CLOCK_SOURCE_NONE	None use clock for FTM
FTM_GPT_IP_CLOCK_SOURCE_SYSTEMCLK	System clock
FTM_GPT_IP_CLOCK_SOURCE_FIXED_FREQUENCY	Fixed frequency
FTM_GPT_IP_CLOCK_SOURCE_EXTERNALCLK	External clock

Definition at line 120 of file `Ftm_Gpt_Ip_Types.h`.

6.1.5.3 Ftm_Gpt_Ip_ChannelModeType

enum `Ftm_Gpt_Ip_ChannelModeType`

Prescaler type. Indicates of whether the clock channel mode is "NORMAL" or "ALTERNATE".

This enumeration specifies the possible types of prescalers used to configure base-clock timers

Channel mode type. Indicates of whether the channel mode is "CONTINUOUS" or "ONE SHOT".

ChannelModeType of channel.

Enumerator

FTM_GPT_IP_CH_MODE_CONTINUOUS	channel mode - continuous mode
FTM_GPT_IP_CH_MODE_ONESHOT	channel mode - one-shot mode.

Definition at line 145 of file `Ftm_Gpt_Ip_Types.h`.

6.2 Gpt Driver

6.2.1 Detailed Description

Macros

- `#define GPT_E_PARAM_CHANNEL`
Function Gpt_StartTimer is called when the driver is in sleep mode for a channel which is not wakeup enabled.
- `#define GPT_E_BUSY`
Function called with parameter value out of range.
- `#define GPT_E_TIMEOUT`
Function called when a timeout is occurred.
- `#define GPT_E_PARAM_CONFIG`
Function called with invalid the parameter in function Gpt_Init.
- `#define GPT_STARTTIMER_ID`
API service ID for Gpt_GetVersionInfo function.
- `#define GPT_PROCESSCOMMONINTERRUPT_ID`
API service ID for Gpt_StopTimer function.
- `#define GPT_INSTANCE_ID`
API service ID for Gpt_ChangeNextTimeoutValue function.
- `#define GPT_VALIDATE_GLOBAL_CALL`
GPT_VALIDATE_GLOBAL_CALL.
- `#define GPT_VALIDATE_CHANNEL_CALL`
GPT_VALIDATE_CHANNEL_CALL.
- `#define GPT_VALIDATE_STATE`
GPT_VALIDATE_STATE.
- `#define GPT_VALIDATE_PARAM`
GPT_VALIDATE_PARAM.

Types Reference

- `typedef uint8 Gpt_ChannelType`
Prescaler type. Indicates of whether the clock channel mode is "GPT_NORMAL" or "GPT_ALTERNATE".

Enum Reference

- `enum Gpt_ModeType`
This enumerated type allows the selection of different power modes.
- `enum Gpt_ChannelModeType`
Gpt channel mode type. Indicates of whether the channel mode is "CONTINUOUS" or "ONE SHOT".

Function Reference

- void [Gpt_Init](#) (const Gpt_ConfigType *configPtr)
GPT driver initialization function.
- void [Gpt_StartTimer](#) ([Gpt_ChannelType](#) channel, Gpt_ValueType value)
GPT driver function for starting a timer channel.
- void [Gpt_StopTimer](#) ([Gpt_ChannelType](#) channel)
GPT driver function for stopping a timer channel.
- void [Gpt_ProcessCommonInterrupt](#) (uint8 channel)
Gpt common handler to implements generic part of the ISR.

6.2.2 Macro Definition Documentation

6.2.2.1 GPT_E_PARAM_CHANNEL

```
#define GPT_E_PARAM_CHANNEL
```

Function [Gpt_StartTimer](#) is called when the driver is in sleep mode for a channel which is not wakeup enabled.

Errors and exceptions that will be detected by the GPT driver.

Function called without module initialization.

Errors and exceptions that will be detected by the GPT driver.

Initialization called when already initialized.

Errors and exceptions that will be detected by the GPT driver.

Function called for invalid channel.

Errors and exceptions that will be detected by the GPT driver.

Definition at line 196 of file Gpt.h.

6.2.2.2 GPT_E_BUSY

```
#define GPT_E_BUSY
```

Function called with parameter value out of range.

Errors and exceptions that will be detected by the GPT driver

Function called when timer channel is still running.

Errors and exceptions that will be detected by the GPT driver.

Definition at line 235 of file Gpt.h.

6.2.2.3 GPT_E_TIMEOUT

```
#define GPT_E_TIMEOUT
```

Function called when a timeout is occurred.

Errors and exceptions that will be detected by the GPT driver.

Definition at line 252 of file Gpt.h.

6.2.2.4 GPT_E_PARAM_CONFIG

```
#define GPT_E_PARAM_CONFIG
```

Function called with invalid the parameter in function Gpt_Init.

Errors and exceptions that will be detected by the GPT driver

API Gpt_SetClockMode service called with wrong parameter.

Parameters used when raising an error/exception

Function called with invalid mode param.

Errors and exceptions that will be detected by the GPT driver

function called for invalid channel on the current core

Errors and exceptions that will be detected by the GPT driver

Definition at line 288 of file Gpt.h.

6.2.2.5 GPT_STARTTIMER_ID

```
#define GPT_STARTTIMER_ID
```

API service ID for Gpt_GetVersionInfo function.

API SERVICE IDs

Parameters used when raising an error/exception

API service ID for Gpt_Init function

Parameters used when raising an error/exception

API service ID for Gpt_DeInit function

Parameters used when raising an error/exception

API service ID for Gpt_GetTimeElapsed function

Parameters used when raising an error/exception

API service ID for Gpt_GetTimeRemaining function

Parameters used when raising an error/exception

API service ID for Gpt_StartTimer function

Parameters used when raising an error/exception

Definition at line 335 of file Gpt.h.

6.2.2.6 GPT_PROCESSCOMMONINTERRUPT_ID

```
#define GPT_PROCESSCOMMONINTERRUPT_ID
```

API service ID for Gpt_StopTimer function.

Parameters used when raising an error/exception

API service ID for Gpt_SetMode function

Parameters used when raising an error/exception

API service ID for Gpt_ProcessCommonInterrupt generic ISR handler

Parameters used when raising an error/exception

Definition at line 396 of file Gpt.h.

6.2.2.7 GPT_INSTANCE_ID

```
#define GPT_INSTANCE_ID
```

API service ID for Gpt_ChangeNextTimeoutValue function.

Parameters used when raising an error/exception

API service ID for Gpt_SetClockMode function

Parameters used when raising an error/exception

API service ID for Gpt_GetPredefTimerValue function

Parameters used when raising an error/exception

API service ID for Gpt_Channel_EnableChainMode function

Parameters used when raising an error/exception

Instance ID of this GPT driver.

Definition at line 435 of file Gpt.h.

6.2.2.8 GPT_VALIDATE_GLOBAL_CALL

```
#define GPT_VALIDATE_GLOBAL_CALL
```

GPT_VALIDATE_GLOBAL_CALL.

Validates the global call uses all the channels - Gpt_Init, Gpt_DeInit, Gpt_SetMode.

Definition at line 72 of file Gpt_EnvCfg.h.

6.2.2.9 GPT_VALIDATE_CHANNEL_CALL

```
#define GPT_VALIDATE_CHANNEL_CALL
```

GPT_VALIDATE_CHANNEL_CALL.

Validates the call for a specific channel.

Definition at line 78 of file Gpt_EnvCfg.h.

6.2.2.10 GPT_VALIDATE_STATE

```
#define GPT_VALIDATE_STATE
```

GPT_VALIDATE_STATE.

Validates the channel status.

Definition at line 84 of file Gpt_EnvCfg.h.

6.2.2.11 GPT_VALIDATE_PARAM

```
#define GPT_VALIDATE_PARAM
```

GPT_VALIDATE_PARAM.

Validates the time value parameter.

Definition at line 90 of file Gpt_EnvCfg.h.

6.2.3 Types Reference

6.2.3.1 Gpt_ChannelType

```
typedef uint8 Gpt_ChannelType
```

Prescaler type. Indicates of whether the clock channel mode is "GPT_NORMAL" or "GPT_ALTERNATE".

This enumeration specifies the possible types of prescalers used to configure base-clock timers

Definition at line 483 of file Gpt.h.

6.2.4 Enum Reference

6.2.4.1 Gpt_ModeType

```
enum Gpt_ModeType
```

This enumerated type allows the selection of different power modes.

Modes of the GPT driver.

Enumerator

GPT_MODE_NORMAL	GPT Normal operation mode of the GPT.
GPT_MODE_SLEEP	GPT Sleep mode.

Definition at line 446 of file Gpt.h.

6.2.4.2 Gpt_ChannelModeType

```
enum Gpt_ChannelModeType
```

Gpt channel mode type. Indicates of whether the channel mode is "CONTINUOUS" or "ONE SHOT".

ChannelModeType of channel.

Enumerator

GPT_CH_MODE_CONTINUOUS	GPT channel mode - continuous mode.
GPT_CH_MODE_ONESHOT	GPT channel mode - one-shot mode.

Definition at line 456 of file Gpt.h.

6.2.5 Function Reference

6.2.5.1 Gpt_Init()

```
void Gpt_Init (
    const Gpt_ConfigType * configPtr )
```

GPT driver initialization function.

This service is a non reentrant function used for driver initialization. The Initialization function shall initialize all relevant registers of the configured hardware with the values of the structure referenced by the parameter ConfigPtr. All time units used within the API services of the GPT driver shall be of the unit ticks. This function shall only initialize the configured resources. Resources that are not configured in the configuration file shall not be touched. The following rules regarding initialization of controller registers shall apply to the GPT Driver implementation: [1] If the hardware allows for only one usage of the register, the driver module implementing that functionality is responsible for initializing the register [2] If the register can affect several hardware modules and if it is an IO register it shall be initialized by the PORT driver [3] If the register can affect several hardware modules and if it is not an IO register it shall be initialized by the MCU driver [4] One-time writable registers that require initialization directly after reset shall be initialized by the startup code [5] All other registers shall be initialized by the startup code

Parameters

in	<i>configPtr</i>	Pointer to a selected configuration structure
----	------------------	---

Returns

void

Precondition

The data structure including the configuration set required for initializing the GPT driver..

6.2.5.2 Gpt_StartTimer()

```
void Gpt_StartTimer (
    Gpt_ChannelType channel,
    Gpt_ValueType value )
```

GPT driver function for starting a timer channel.

The function Gpt_StartTimer shall start the selected timer channel with a defined time-out period. The function Gpt_StartTimer shall invoke the configured notification for that channel (see also GPT292) after the time-out period referenced via the parameter value (if enabled). All time units used within the API services of the GPT driver shall be of the unit ticks. In production mode no error is generated. The rational is that it adds no additional functionality to the driver. In this case the timer will be restarted with the time-out value, given as a parameter to the service. Usage of re-entrant capability is only allowed if the callers take care that there is no simultaneous usage of the same channel. To get times out of register values it is necessary to know the oscillator frequency, pre-scalers and so on. Since these settings are made in MCU and(or) in other modules it is not possible to calculate such times. Hence the conversions between time and ticks shall be part of an upper layer. The driver needs to be initialized before calling [Gpt_StartTimer\(\)](#). Otherwise, the function Gpt_StartTimer shall raise the development error GPT_E_UNINIT.

Module Documentation

Parameters

in	<i>channel</i>	channel id
in	<i>value</i>	time-out period (in number of ticks) after a notification or a wakeup event shall occur.

Returns

void

Precondition

The driver needs to be initialized.

6.2.5.3 Gpt_StopTimer()

```
void Gpt_StopTimer (
    Gpt_ChannelType channel )
```

GPT driver function for stopping a timer channel.

Service for stopping the selected timer channel Stopping a timer channel, not been started before will not return a development error Timer channels configured in one shot mode are stopped automatically, when the time-out period has expired. Usage of re-entrant capability is only allowed if the callers take care that there is no simultaneous usage of the same channel. The driver needs to be initialized before calling [Gpt_StopTimer\(\)](#). Otherwise, the function shall raise the development error GPT_E_UNINIT.

Parameters

in	<i>channel</i>	channel id
----	----------------	------------

Returns

void

Precondition

The driver needs to be initialized. Gpt_StartTimer must be called before.

6.2.5.4 Gpt_ProcessCommonInterrupt()

```
void Gpt_ProcessCommonInterrupt (
    uint8 channel )
```

Gpt common handler to implements generic part of the ISR.

Generic function used by all interrupt service routines to call notification functions and wakeup the EcuM

Module Documentation

Parameters

in	<i>channel</i>	logic channel number
----	----------------	----------------------

Returns

void

Precondition

The driver needs to be initialized.

6.3 Pit IPL

6.3.1 Detailed Description

Enum Reference

- enum [Pit_Ip_StatusType](#)
Pit Status error.
- enum [Pit_Ip_ChannelModeType](#)
Channel mode type. Indicates of whether the channel mode is "CONTINUOUS" or "ONE SHOT".

6.3.2 Enum Reference

6.3.2.1 Pit_Ip_StatusType

enum [Pit_Ip_StatusType](#)

Pit Status error.

Status error

Enumerator

PIT_IP_SUCCESS	Status value is SUCCESS.
PIT_IP_ERROR	Status value is ERROR

Definition at line 107 of file Pit_Ip_Types.h.

6.3.2.2 Pit_Ip_ChannelModeType

enum [Pit_Ip_ChannelModeType](#)

Channel mode type. Indicates of whether the channel mode is "CONTINUOUS" or "ONE SHOT".

ChannelModeType of channel.

Enumerator

PIT_IP_CH_MODE_CONTINUOUS	hannel mode - continuous mode
PIT_IP_CH_MODE_ONESHOT	hannel mode - one-shot mode.

Module Documentation

Definition at line 117 of file Pit_Ip_Types.h.

6.4 PIT IPL

6.4.1 Detailed Description

Macros

- `#define GPT_STOP_SEC_CODE`
Pit_Ip_SetUserAccessAllowed.

6.4.2 Macro Definition Documentation

6.4.2.1 GPT_STOP_SEC_CODE

```
#define GPT_STOP_SEC_CODE
```

Pit_Ip_SetUserAccessAllowed.

This function is called externally by OS Application. note: with S32ZE platform only PIT_AE support usermode, so user can only pass base address of PIT_AE.

Parameters

in	<i>PitBaseAddr</i>	- The base address of PIT.
----	--------------------	----------------------------

Definition at line 117 of file *Pit_Ip_TrustedFunctions.h*.

6.5 Rtc IPL

6.5.1 Detailed Description

Enum Reference

- enum [Rtc_Ip_ClockSelectType](#)
Enum containing the RTC module clock sources.
- enum [Rtc_Ip_InterruptType](#)
Enum containing RTC interrupt type.
- enum [Rtc_Ip_ModeType](#)
Enum containing RTC interrupt mode.
- enum [Rtc_Ip_StatusType](#)
Rtc Status error.
- enum [Rtc_Ip_ChannelModeType](#)
Channel mode type. Indicates of whether the channel mode is "CONTINUOUS" or "ONE SHOT".

6.5.2 Enum Reference

6.5.2.1 Rtc_Ip_ClockSelectType

enum [Rtc_Ip_ClockSelectType](#)

Enum containing the RTC module clock sources.

Rtc_Ip_ClockSelectType

Enumerator

RTC_IP_CLOCK_SOURCE↔ _0	RTC clock source 0.
RTC_IP_CLOCK_SOURCE↔ _1	RTC clock source 1.
RTC_IP_CLOCK_SOURCE↔ _2	RTC clock source 2.
RTC_IP_CLOCK_SOURCE↔ _3	RTC clock source 3.

Definition at line 121 of file Rtc_Ip_Types.h.

6.5.2.2 Rtc_Ip_InterruptType

enum [Rtc_Ip_InterruptType](#)

Enum containing RTC interrupt type.

Rtc_Ip_InterruptType

Enumerator

RTC_IP_COUNTER_INTERRUPT	RTC_COUNTER_INTERRUPT
RTC_IP_API_INTERRUPT	RTC_API_INTERRUPT
RTC_IP_ROLLOVER_INTERRUPT	RTC_ROLLOVER_INTERRUPT.

Definition at line 134 of file Rtc_Ip_Types.h.

6.5.2.3 Rtc_Ip_ModeType

```
enum Rtc_Ip_ModeType
```

Enum containing RTC interrupt mode.

Rtc_Ip_ModeType

Enumerator

RTC_IP_API_MODE	API(Autonomous periodic interrupt) Mode
RTC_IP_RTC_MODE	RTC Mode

Definition at line 146 of file Rtc_Ip_Types.h.

6.5.2.4 Rtc_Ip_StatusType

```
enum Rtc_Ip_StatusType
```

Rtc Status error.

Status error

Enumerator

RTC_IP_SUCCESS	Status value is SUCCESS.
RTC_IP_ERROR	Status value is ERROR

Definition at line 157 of file Rtc_Ip_Types.h.

6.5.2.5 Rtc_Ip_ChannelModeType

enum [Rtc_Ip_ChannelModeType](#)

Channel mode type. Indicates of whether the channel mode is "CONTINUOUS" or "ONE SHOT".

ChannelModeType of channel.

Enumerator

RTC_IP_CH_MODE_CONTINUOUS	channel mode - continuous mode
RTC_IP_CH_MODE_ONESHOT	channel mode - one-shot mode.

Definition at line 168 of file Rtc_Ip_Types.h.

6.6 Stm IPL

6.6.1 Detailed Description

Macros

- `#define GPT_STOP_SEC_CODE`
Stm_Ip_SetUserAccessAllowed.

Enum Reference

- enum `Stm_Ip_ChannelModeType`
Channel mode type. Indicates of whether the channel mode is "CONTINUOUS" or "ONE SHOT".

6.6.2 Macro Definition Documentation

6.6.2.1 GPT_STOP_SEC_CODE

```
#define GPT_STOP_SEC_CODE
```

`Stm_Ip_SetUserAccessAllowed.`

This function is called externally by OS Application

Parameters

in	<i>StmBaseAddr</i>	- The base address of Stm.
----	--------------------	----------------------------

Definition at line 117 of file `Stm_Ip_TrustedFunctions.h`.

6.6.3 Enum Reference

6.6.3.1 Stm_Ip_ChannelModeType

```
enum Stm_Ip_ChannelModeType
```

Channel mode type. Indicates of whether the channel mode is "CONTINUOUS" or "ONE SHOT".

ChannelModeType of channel.

Enumerator

STM_IP_CH_MODE_CONTINUOUS	channel mode - continuous mode
STM_IP_CH_MODE_ONESHOT	channel mode - one-shot mode.

Definition at line 119 of file Stm_Ip_Types.h.

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